LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 1

Knowledge:

Multiple Choice Questions:

B **What is the primary purpose of an electronic communication system?**

A) To amplify signals for long-distance transmission
B) To produce an accurate replica of the transmitted information
C) To modify the original signal for security purposes
D) To store data for later use

D **Which of the following is NOT a form of information transmitted in communication systems?**

A) Speech
B) Video
C) Data
D) Heat

B **What is the function of a transducer in a communication system?**

A) To amplify signals 把一种形式的能量转换成另一种
B) To convert one form of energy into another
C) To store data for future use
D) To reduce noise in a channel

C **Which of the following describes a 'simplex' communication system?**

A) It allows two-way communication simultaneously
B) It allows two-way communication but not at the same time
C) It allows communication in only one direction
D) It requires modulation for operation

B **Which medium is used in a guided propagation communication channel?**

A) Radio waves
B) Fiber-optic cables
C) Infrared signals

D) Ultrasonic waves

A

What is 'attenuation' in a communication channel?

A) Loss of energy in the signal
B) Distortion of the signal
C) Noise interference in the signal
D) Delay in the signal transmission

C

In communication systems, what does SNR (Signal-to-Noise Ratio) indicate?

A) The amount of distortion in the signal
B) The bandwidth required for signal transmission
C) The strength of the signal compared to noise
D) The time delay in the signal

B

Which of the following is a characteristic of optical fibre communication?

A) High signal attenuation
B) High bandwidth and low interference
C) High susceptibility to electromagnetic interference
D) Large size and heavy weight

C

What type of signal is typically transmitted in wireless broadcast channels like TV and radio?

A) Unmodulated signals
B) Baseband signals
C) Modulated signals
D) Amplified signals

D

Which component in a communication system is responsible for converting the received signal back to its original form?

A) Transmitter
B) Channel
C) Modulator
D) Receiver

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Understanding:

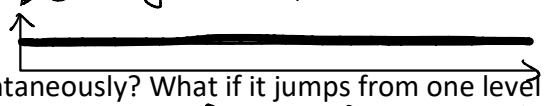
1. Identify the key features of analogue signals.
   Continuous,
2. What is the primary function of a communication system?
   Transmit information from source to information.
3. What role does a transducer play in a communication system?
   Message → Electric signal
4. What is the difference between simplex, half-duplex, and full-duplex communication systems?

5. What is attenuation, and how does it affect communication signals?
   Loss of energy
6. How is noise defined in a communication channel, and what is its effect on transmitted signals?

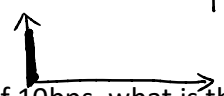
7. What is the relationship between period and frequency? Give three numerical examples with your explanation.
   $f = \frac{1}{T}$, $T = \frac{1}{f}$ 100 Hz ⇒ 0.01 s
8. What is the Morse code for Lancaster? Try to write Morse code for your name.

A ● −	J ● − − −	S ●
B − ● ● ●	K − ● −	T ·
C − ● − ●	L ● − ● ●	U
D − ● ●	M − −	V
E ●	N − ●	W
F ● ● − ●	O − − −	X
G − − ●	P ● − − ●	Y −
H ● ● ● ●	Q − − ● −	Z ·

9. What if a signal maintains a constant amplitude for the entire active time? In such a case, what is the frequency? DC signal, f = 0



10. What if a signal changes instantaneously? What if it jumps from one level to another in no time? What is the frequency? f = infinite T = 0



11. A digital signal has a bit rate of 10bps, what is the duration of each bit (the bit interval)? Clue: 1000 ms = 1 second 10 bps, 0.1 s = 100 ms